

Somaiya Vidyavihar University



WEB PROGRAMMING LABORATORY Mini Project Report-A

Project Title:

Hotel Management System - WPL Project

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Department of Computer Engineering

Declaration

This is to certify that the mini project titled **Hotel Management System** is a bonafide work carried out by the undersigned students under the guidance of the course instructors as part of the Web Programming Laboratory during the academic year 2025–2026.

Student Signature(s): _____

Date: 30-03-2026

Acknowledgement

We express our sincere gratitude to the Department of Computer Engineering, K. J. Somaiya School of Engineering, Somaiya Vidyavihar University for providing the opportunity to work on this mini project. We would like to thank our course instructor Dr. Pradnya Gotmare for their guidance and support.

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1. Introduction

The Hotel Management System is a web-based application designed to provide hotel booking and management services similar to platforms like Agoda, Booking.com, and Airbnb. It allows users to search for available rooms, view detailed information, compare prices, and make bookings online in a simple and efficient manner. Traditional hotel management systems rely on manual processes, which are time-consuming, less efficient, and prone to human errors. This system aims to overcome these limitations by digitizing all core operations such as room availability tracking, customer data management, and billing.

The application provides a centralized platform for both customers and hotel administrators, ensuring smooth communication and data handling. Users can easily browse rooms, check availability in real time, and confirm bookings instantly. The system is developed using modern web technologies such as HTML, CSS, JavaScript, PHP, and MySQL, making it reliable and easy to maintain. It also focuses on providing a user-friendly interface that enhances the overall user experience. By automating hotel operations and enabling online booking, the system helps improve efficiency, reduce workload, and increase customer satisfaction. Overall, it serves as a scalable and practical solution for modern hotel management needs.

2. Problem Statement

The hospitality industry requires efficient management of room bookings, customer data, and billing processes to ensure smooth operations and high customer satisfaction. However, many hotels still rely on traditional or semi-manual systems that are time-consuming, error-prone, and inefficient in handling large volumes of data. These systems make it difficult to track room availability in real time, manage customer records accurately, and generate bills without errors.

Additionally, unlike modern online platforms such as Agoda and Booking.com, smaller hotels often lack a centralized system that allows customers to easily browse, compare, and book rooms online. This results in poor customer experience, limited accessibility, and loss of potential revenue.

Therefore, there is a need for a web-based Hotel Management System that can automate hotel operations, provide real-time room availability, enable online booking, and

maintain accurate digital records. The system should be user-friendly, efficient, and capable of improving both operational performance and customer satisfaction.

3. Objectives

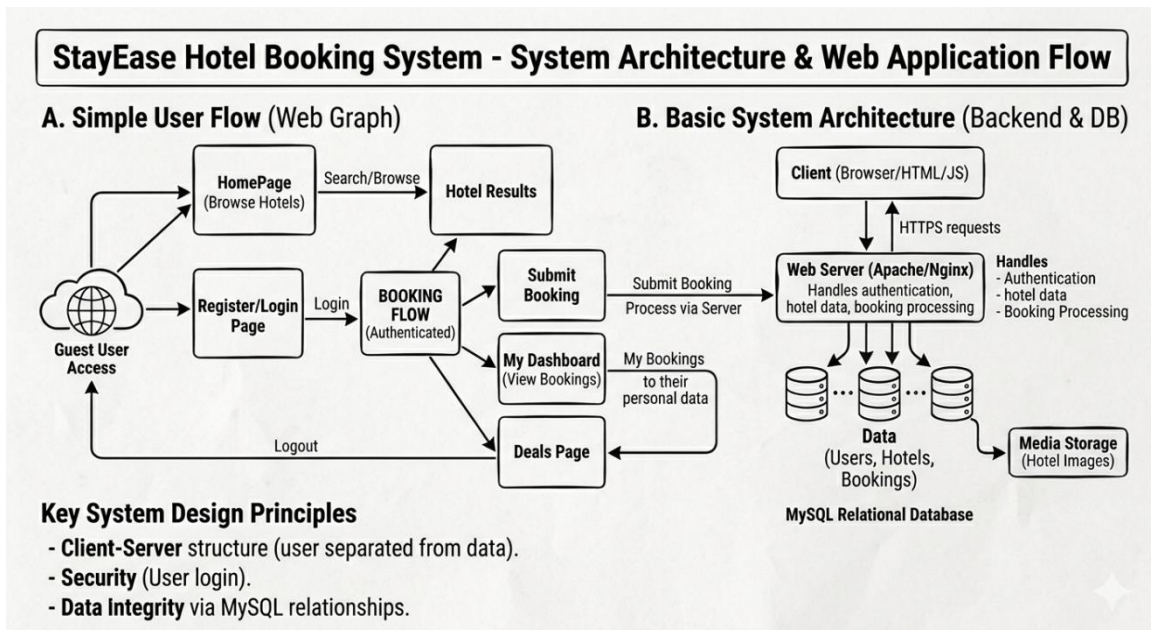
The main objectives of the Hotel Management System project are as follows:

- To develop a web-based application that simplifies hotel booking and management processes.
- To provide an interface for users to view available rooms and make bookings online.
- To maintain and manage customer information in a structured and secure database.
- To enable administrators to add, update, and manage room details and availability.
- To automate the booking process and reduce manual work involved in hotel operations.
- To provide real-time updates on room availability and booking status.
- To implement a system for handling check-in, check-out, and billing processes efficiently.
- To design a user-friendly interface for both customers and administrators.
- To ensure data accuracy and reduce errors associated with manual record keeping.
- To create a scalable system that can be extended with additional features in the future.

4. System Architecture / Web Graph

Home → Signup → Login → Room Search → Booking → Customer Details → Billing → Logout

The user first lands on the Home page, where they can navigate to the Signup page to create a new account. After successfully signing up, the user proceeds to the Login page for authentication. Once logged in, the user can search for available rooms using the Room Search feature. If the user finds a suitable room, they can proceed to the Booking page to reserve it. Next, the user fills in their Customer Details for the booking confirmation and billing process. After completing the transaction, the user can logout securely. This flow ensures a smooth and efficient experience for users to browse, book, and manage hotel stays online



5. Web Pages Developed

Page Name	Description
Home Page	Displays introduction and navigation menu for the hotel booking website.
Login Page	Allows users to log in using their credentials for authentication.
Signup Page	Allows new users to create an account on the platform.
Hotel Listing Page	Displays the available hotels along with their basic details and images.
Hotel Details / Booking Page	Shows detailed information about selected hotels and provides booking options.
Contact Page	Allows users to contact the company through the details provided.
Hotel Details / Booking Page	Shows users to contact the company through the details provided.
Contact Page	Allows users to contact the company through the details provided.

CSS Files

- auth.css
- bg.css

- booking.css
- contact.css
- homepage.css
- hotel.css

JavaScript Files

- auth.js

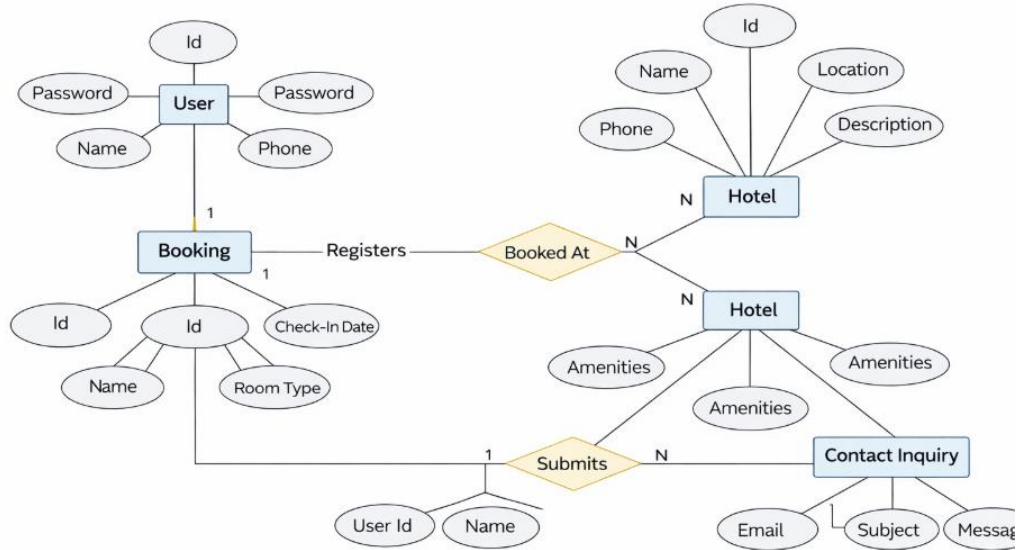
PHP Files

- booking_process.php
- bookings.php
- dashboard.php
- homepage.php
- hotel.php
- login_process.php
- logout.php
- register_process.php

HTML Files

- contact.html
- login.html
- signup.html

6. Database Design



RELATIONAL MODEL SCHEMA –

1. User

Field	Type	Description
user_id	INT(PK)	Unique ID for each user
name	VARCHAR	Name of the user
email	VARCHAR	Email of the user (unique)
password	VARCHAR	Encrypted password
phone	BIGINT	Contact number of user

2. Hotel

Field	Type	Description
hotel_id	INT(PK)	Unique ID for each hotel
hotel_name	VARCHAR	Name of the hotel
location	VARCHAR	Location of the hotel
price	INT	Price per night
image	VARCHAR	Hotel image path or URL
description	TEXT	Detailed description of the hotel
rating	FLOAT	Rating of the hotel

3. Booking

Field	Type	Description
booking_id	INT(PK)	Unique ID for each booking
user_id	INT(FK)	References the user making the booking
hotel_id	INT(FK)	References the booked hotel
check_in_date	DATE	Check-in date of booking
check_out_date	DATE	Check-out date of booking
guests	INT	Number of guests
total_price	INT	Total booking amount
booking_status	VARCHAR	Status of booking (Confirmed / Pending / Cancelled)

4. Contact

Field	Type	Description
contact_id	INT(PK)	Unique ID for each contact message
user_id	INT(FK)	References the user sending the message
name	VARCHAR	Name of the sender
email	VARCHAR	Email of the sender
subject	VARCHAR	Subject of the message
message	TEXT	Message or inquiry submitted by the user

7. GitHub Repository

<https://github.com/dev-coder-1/Hotel-Management-WPL-PROJECT.git>

8. References

- MDN Web Docs
<https://developer.mozilla.org/en-US/>

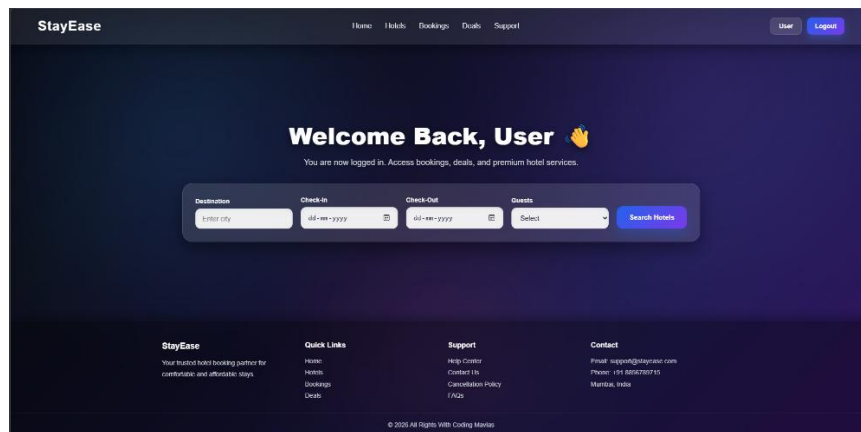
- **HTML Reference – MDN**
<https://developer.mozilla.org/en-US/docs/Web/HTML>
- **CSS Reference – MDN**
<https://developer.mozilla.org/en-US/docs/Web/CSS>
- **JavaScript Guide – MDN**
<https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>
- **PHP MySQLi Manual**
<https://www.php.net/manual/en/book.mysqli.php>

9.Future Scope

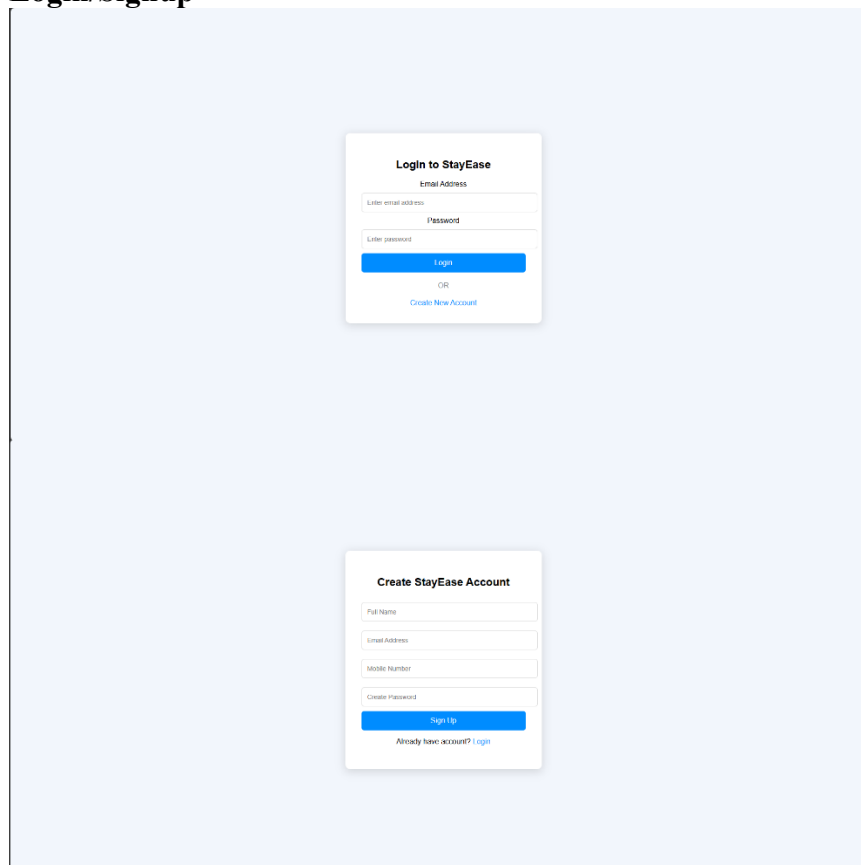
- **Online Payment Integration**
Secure online payment methods can be added for easier hotel booking.
- **Room Availability Management**
The project can be upgraded to display real-time room availability and prevent double bookings.
- **Admin Dashboard**
An admin panel can be developed to manage hotels, users, and bookings. .
- **Reviews and Ratings System**
Users can be allowed to give feedback and ratings for hotels.
- **Mobile App and Notifications**
The project can be expanded into a mobile app with booking alerts and notifications

10. Appendix (Source Code / Screenshots)

- Home Page



- **Login/Signup**



- **Booking Page**



The Taj Palace Mumbai
Mumbai

The Taj Palace Mumbai

★ 4.8

- ✓ King Size Bed
- ✓ Free WiFi
- ✓ AC Room
- ✓ Breakfast Included
- ✓ 2 Guests

Price per night

₹7,500 / night

Book Your Stay

Fill in your details to confirm your booking

Full Name

Vedant

Email

vedant@gmail.com

Phone Number

Enter your phone number

Check-in Date

dd-mm-yyyy

Check-out Date

dd-mm-yyyy

Guests

1

Room Type

Deluxe Room

Special Request

Any special request...

Confirm Booking

Explore Hotels

Discover top-rated stays, compare prices, and book your perfect hotel.

Search by city or hotel name...

All Prices

All Ratings



The Taj Palace Mumbai
Mumbai

★ 4.8

₹7,500 / night

Book Now



Marine Drive Residency
Mumbai

★ 4.5

₹5,200 / night

Book Now



Pune Grand Stay
Pune

★ 4.6

₹4,800 / night

Book Now

- **Contact US**

Contact Us

We're here to help you with bookings, support, and any hotel-related queries.

Get In Touch

Reach out to our support team anytime. We are available 24/7 for your travel needs.

Email

support@stayease.com

Phone

+91 98765 43210

Address

Mumbai, Maharashtra, India

Working Hours

24/7 Customer Support

Send a Message

Full Name

Enter your full name

Email Address

Enter your email

Subject

Enter subject

Message

Write your message here...

Send Message

